

Python implementation of the Moore1997 Loudness model

The base code that inspired this project is in the AMT toolbox : <https://amtoolbox.org/>

The current state of the project :

The Moore1997 model is implemented and is being tested, it is namely compared to the results that are displayed in the original article.

This document is a description of the code base with a few notable results.

Code Base :

Firstly, this project only implements one model but the code is made to be potentially adaptable to other model of the amt toolbox, this is why the algorithms are separated into modules.

- The model file contains the Moore1997 model, it is the same code as the amt-toolbox one with the pretreatment (independent of the input signal) done in the constructor.

No complexity enhancement were done yet.

- The data that is used by this model is in the dataPreparator file, performing the same treatment as Glasbergdata2002 in the amt toolbox.

- Finally, all computations are done in separate files, such as FFT, Interpolation etc.

Note that a numerical error report is available to compare the implementation of several features of the toolbox.

To run the model, you can create a model :

Model :

Arguments : Free field : boolean

Then run moore1997 :

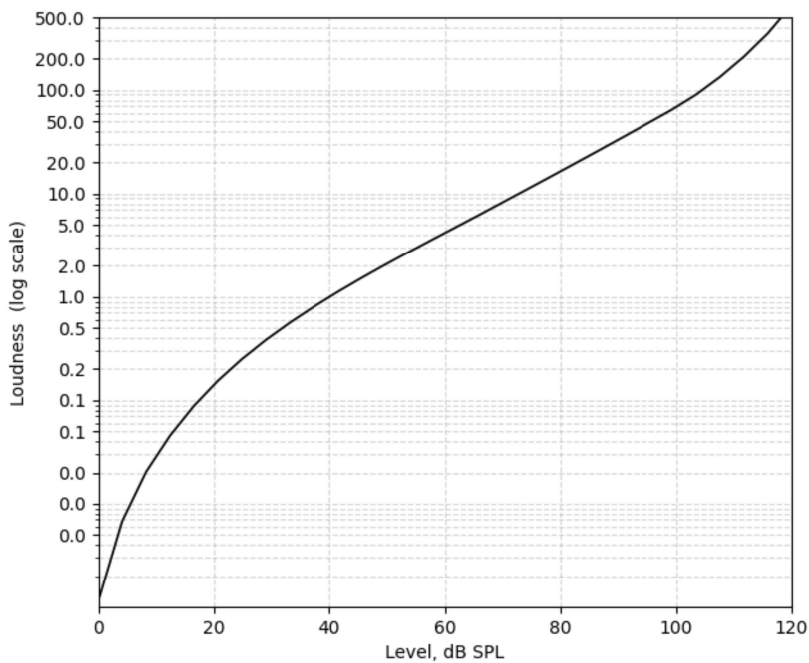
Arguments :

-earSig : the signal

To obtain some of the figures that are found in the article, you can use the experiment file.

Fig12Plot :

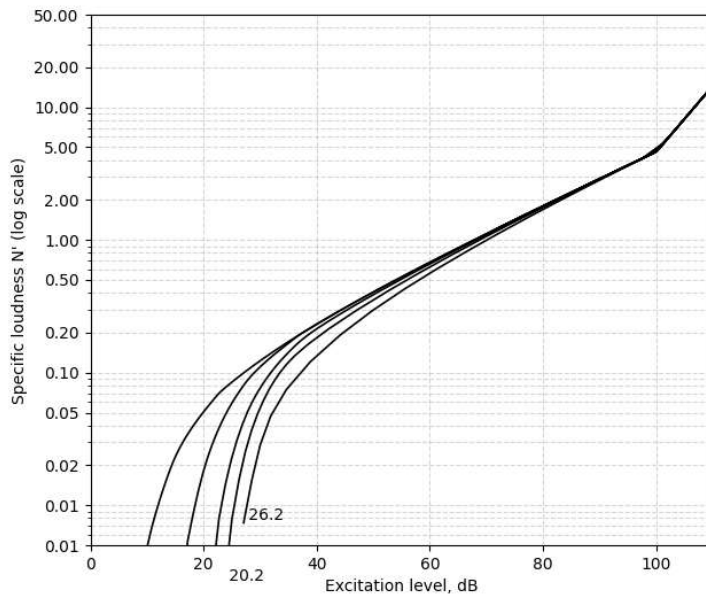
Arguments : Frequency of the generated signals with which the curve will be computed, use 1kHz to fit the model's fig12 curve.



This is the curve generated by 1kHz sinusoids.

Fig8Plot :

Arguments : Level in dB at which signals will be generated to plot the specific loudness.



This is the graph generated for signals at 110 dB.

EldbPlot : No Arguments : Reproduces the figure 5 from the paper.

